

System Binary Proxy Execution

Sub-techniques (14)

Adversaries may bypass process and/or signature-based defenses by proxying execution of malicious content with signed, or otherwise trusted, binaries. Binaries used in this technique are often Microsoft-signed files, indicating that they have been either downloaded from Microsoft or are already native in the operating system.^[1] Binaries signed with trusted digital certificates can typically execute on Windows systems protected by digital signature validation. Several Microsoft signed binaries that are default on Windows installations can be used to proxy execution of other files or commands.

Similarly, on Linux systems adversaries may abuse trusted binaries such as `split` to proxy execution of malicious commands.^{[2][3]}

ID: T1218

Sub-techniques: [T1218.001](#), [T1218.002](#), [T1218.003](#), [T1218.004](#), [T1218.005](#), [T1218.007](#), [T1218.008](#), [T1218.009](#), [T1218.010](#), [T1218.011](#), [T1218.012](#), [T1218.013](#), [T1218.014](#), [T1218.015](#)

(i)

Tactic: [Stealth](#)

(i)

Platforms: Linux, Windows, macOS

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[Version Permalink](#)

Procedure Examples

ID	Name	Description
G0032	Lazarus Group	Lazarus Group lnk files used for persistence have abused the Windows Update Client (<code>wuauclt.exe</code>) to execute a malicious DLL. ^{[4][5]}
G1017	Volt Typhoon	Volt Typhoon has used native tools and processes including living off the land binaries or "LOLBins" to maintain and expand access to the victim networks. ^[6]

Mitigations

ID	Mitigation	Description
M1042	Disable or Remove Feature or Program	Many native binaries may not be necessary within a given environment.
M1038	Execution Prevention	Consider using application control to prevent execution of binaries that are susceptible to abuse and not required for a given system or

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		network.
M1050	Exploit Protection	Microsoft's Enhanced Mitigation Experience Toolkit (EMET) Attack Surface Reduction (ASR) feature can be used to block methods of using using trusted binaries to bypass application control.
M1037	Filter Network Traffic	Use network appliances to filter ingress or egress traffic and perform protocol-based filtering. Configure software on endpoints to filter network traffic.
M1026	Privileged Account Management	Restrict execution of particularly vulnerable binaries to privileged accounts or groups that need to use it to lessen the opportunities for malicious usage.
M1021	Restrict Web-Based Content	Restrict use of certain websites, block downloads/attachments, block Javascript, restrict browser extensions, etc.

Detection Strategy

ID	Name	Analytic ID	Analytic Description
		AN0226	Execution of trusted, Microsoft-signed binaries such as <code>rundll32.exe</code> , <code>msiexec.exe</code> , or <code>regsvr32.exe</code> used to execute externally hosted, unsigned, or suspicious payloads through command-line parameters or network retrieval.
DET0081	Detection of Proxy Execution via Trusted Signed Binaries Across Platforms	AN0227	Execution of trusted system binaries (e.g., <code>split</code> , <code>tee</code> , <code>bash</code> , <code>env</code>) used in uncommon sequences or chained behaviors to execute malicious payloads or perform actions inconsistent with normal system or script behavior.
		AN0228	Use of system binaries such as <code>osascript</code> , <code>bash</code> , or <code>curl</code> to download or execute unsigned code or files in conjunction with application proxying.

References

1. [Oddvar Moe et al. \(2022, February\). Living Off The Land Binaries, Scripts and Libraries. Retrieved March 7, 2022.](#)
2. [Torbjorn Granlund, Richard M. Stallman. \(2020, March null\). split\(1\) — Linux manual page. Retrieved March 25, 2022.](#)
3. [GTFOBins. \(2020, November 13\). split. Retrieved April 18, 2022.](#)
4. [Saini, A. and Hossein, J. \(2022, January 27\). North Korea's Lazarus APT leverages Windows Update client, GitHub in latest campaign. Retrieved January 27, 2022.](#)

5. [Pradhan, A. \(2022, February 8\). LolZarus: Lazarus Group Incorporating Lolbins into Campaigns. Retrieved March 22, 2022.](#)
6. [CISA et al.. \(2024, February 7\). PRC State-Sponsored Actors Compromise and Maintain Persistent Access to U.S. Critical Infrastructure. Retrieved May 15, 2024.](#)